MIL-I-8404A(USAF)
15 Sep 1967
Superseding
MIL-I-8404(USAF)
15 June 1953

MILITARY SPECIFICATION

INDICATOR, PRESSURE, SINGLE, 0 TO 35 PSI FUEL PRESSURE TYPE C-33

1. SCOPE

1.1 This specification covers one type of fuel pressure indicator, designated Type C-33.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-F-6836	Flanges, Mounting, Attachable, Aircraft Instrument
MIL-G-8402	Gage, Pressure, Dial Indicating, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-834	Packaging Data Forms, Instructions for Preparation and Use of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33649	Bosses, Fluid Connection-Internal Straight Thread

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The Type C-33 pressure indicator furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 General. The requirements of MIL-G-8402 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 Conflict of requirements. In the event the requirements of MIL-G-8402 and this specification conflict, the requirements of this specification shall govern.

3.3 Design and construction.

*3.3.1 Case. The case shall conform to Figure 1 and to MS33639. The case length shall be 1-7/8 ± 1/8 inch. The case shall be rain-tight and shall incorporate a small vent hole in the bottom.

3.3.2 Pointer. The pointer shall be in accordance with MS33585, except that the length of the pointer shall be such that the tip will extend into the scale a distance equal to 1/3 to 2/3 the length of the shortest graduation.

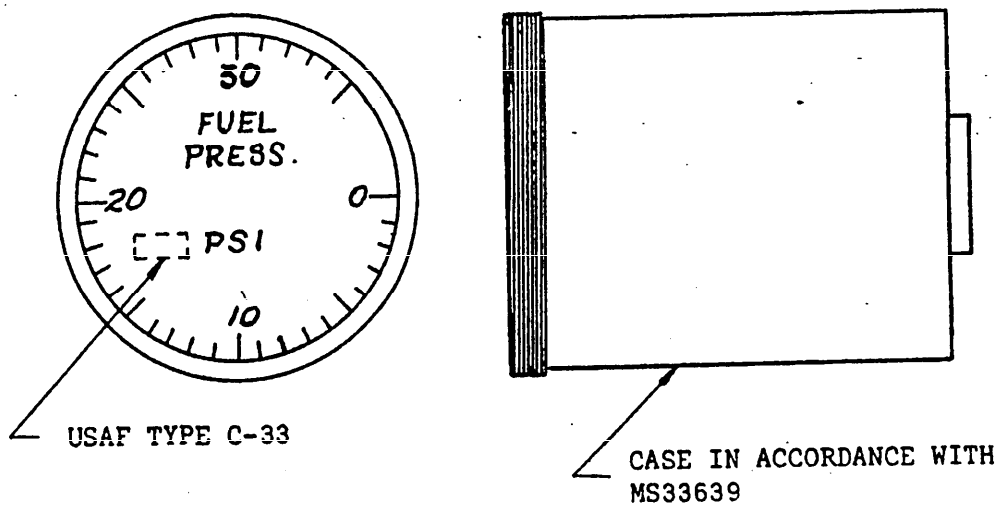
3.3.2.1 Pointer finish. The visible position of the pointer shall be finished in fluorescent-luminescent material.

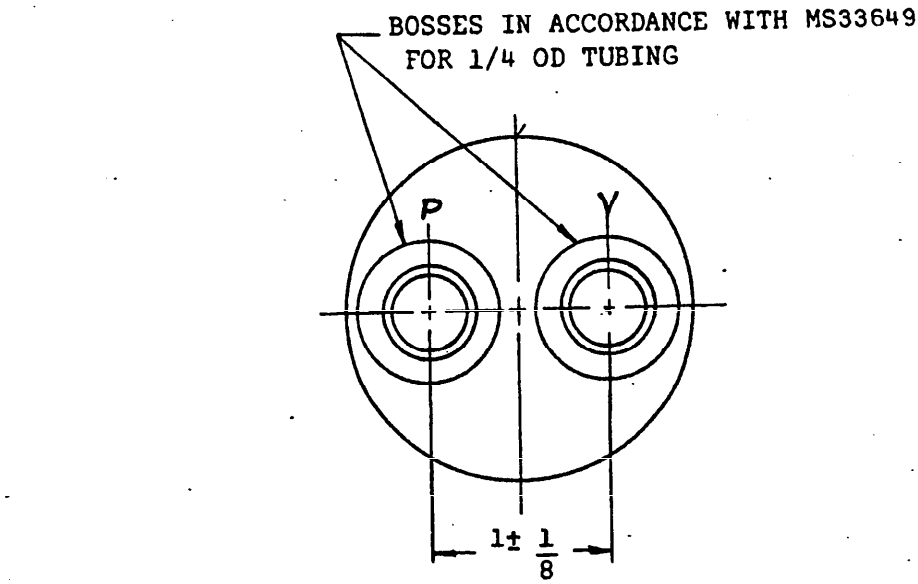
3.3.2.2 The pointer and linkage mechanism of the gage shall be sealed from fuel or vapor entering through the pressure and vent connections.

3.3.3 Dial. The dial shall conform to Figure 1. The 20-psi graduation shall be located on the horizontal center line at the left side of the dial.

3.3.3.1 Dial finish. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

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REAR VIEW

DIMENSIONS IN INCHES.

FIGURE 1. Case and Dial

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	Height or Length Inch ± 1/64	Width of Line or Graduation Inch ± 0.005
Numerals 0, 10, 20, 30	5/32	0.025
5-pound graduations	3/16	0.031
One pound graduations	3/32	0.020
Lettering "FUEL PRESS"	1/8	0.025
Lettering "PSI"	1/8	0.025

3.3.3.2 Durable dull black. The following markings shall be permanently and legibly marked on the dial. These markings and all other markings not otherwise specified shall be finished in durable dull black. The height of the markings shall be as follows:

	Height, Inch
Lettering "USAF"	1/16
Lettering "TYPE C-33"	1/16

3.3.3.2.1 The location of the markings "USAF" and "TYPE C-33" on the dial is optional.

3.4 Weight. The weight of the indicator shall not exceed 0.50 pounds.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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*4.2 General. The indicator shall be subjected to the inspection and testing specified in MIL-G-8402, except as specified herein.

4.3 Performance. The performance of the indicator shall be checked against Tables I and II of this specification.

4.4 Qualification tests. Qualification tests shall consist of all the tests of MIL-G-8402 and those specified herein.

TABLE I

Test Pressure	Scale Error Tolerance ± PSI		Friction Error Tolerance ± PSI	
	Room Temperature	High and Low Temperature	Room Temperature	High and Low Temperature
0	0.2	0.3		
5	0.3	0.4	0.3	0.5
10	0.4	0.6		
15	0.5	0.6	0.3	0.5
20	0.5	0.8		
25	0.5	0.8	0.3	0.5
30	0.6	0.9		
35	0.6	0.9	0.3	0.5

TABLE II

Test	Test Pressure PSI	Tolerance ± PSI
Position error	20	0.2
Vibration	20	
Pointer oscillation		0.3
Pointer variation		0.3
Over-pressure	50	0.3

*4.4.1 Differential pressure error. The gage shall be tested for scale errors with a pressure of 15 psi applied to the vent connection. The test shall be made by subjecting the instrument to the pressures specified in Table III to produce the gage readings shown. The scale errors shall not exceed ± 3 percent of full scale reading.

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TABLE III

Test Pressure (PSI)		Gage Indicator (PSI)
Vent Connection	Pressure Connection	
15	15	0
15	20	5
15	25	10
15	30	15
15	35	20
15	40	25
15	45	30
15	50	35

4.5 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4, MIL-P-116, to insure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Each indicator shall be preserved and packaged in quantity unit packs of one each in accordance with Level "A" requirements as defined in MIL-STD-794, unless otherwise specified in the contract or order. In the event detailed preservation and packaging instructions are not provided in the contract or order, the appropriate preservation method shall be selected from the preservation selection charts contained within MIL-STD-834. Accomplishment of the selected method shall be in accordance with MIL-P-116.

5.2 Packing. Packing for shipment shall be in accordance with Level "B" requirements as defined within MIL-STD-794, unless otherwise specified in the contract or order.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator, Pressure, Single, 0 to 35 PSI Fuel Pressure Type C-33
Specification MIL-I-8404
Mfr's Part No. *

*Applicable data to be entered by the contractor

6. NOTES

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6.1 Intended use. The Type C-33 fuel pressure indicator covered by this specification is intended for use to indicate directly the pressure in the main fuel system of the aircraft engine.

6.2 Alternate use. The Type C-33 fuel pressure indicator can be used as a replacement of the Type C-32 by the addition of an instrument mounting bezel in accordance with MIL-F-6836.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable preparation for shipment instructions (see 5.1 and 5.2).

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List, is Oklahoma City Air Materiel Area (OCNE), Tinker AFB, Oklahoma, and information pertaining to qualification of products may be obtained from that activity.

*6.5 Change indication. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodian:
Air Force - 71

Review activity:
Air Force - 71

Preparing activity:
Air Force - 71

Project Number 6620-F006

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Air Force OC, 24 Dec 64-500

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DEPARTMENT OF THE AIR FORCE

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MIL-I-8404A (USAF)
AMENDMENT 1
28 October 1970

MILITARY SPECIFICATION

INDICATOR, PRESSURE, SINGLE, 0 TO 35 PSI
FUEL PRESSURE TYPE C-33

This amendment forms a part of Military Specification MIL-I-8404 (USAF) dated 15 September 1967.

Page 2, paragraph 3.3.1: Delete entire paragraph and substitute the following: "The case shall conform to figure 1 and MS33639 with a maximum length of 2 inches and shall be rain tight.

Custodian
Air Force - 71

Preparing activity:
Air Force - 71

Reviewing Activity:
Air Force - 71

Project Number:
6620-F052

FSC: 6620

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